CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 15

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Recap of Previous Lecture

Topic Matching Set Part-02

Topics to be Covered

Topic Planarity

Planarity:

Graph

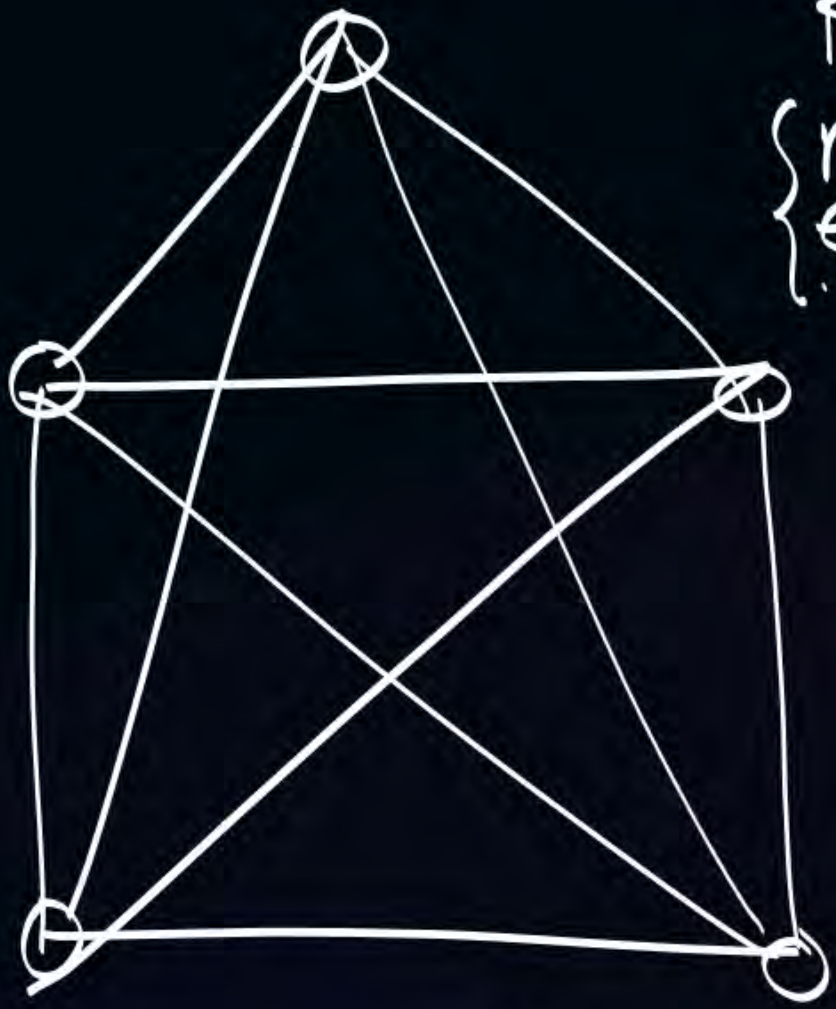
Planar:

if we can draw graph on a plane without intersection of its edges

nonplanar:

otherwise it is nonplanar

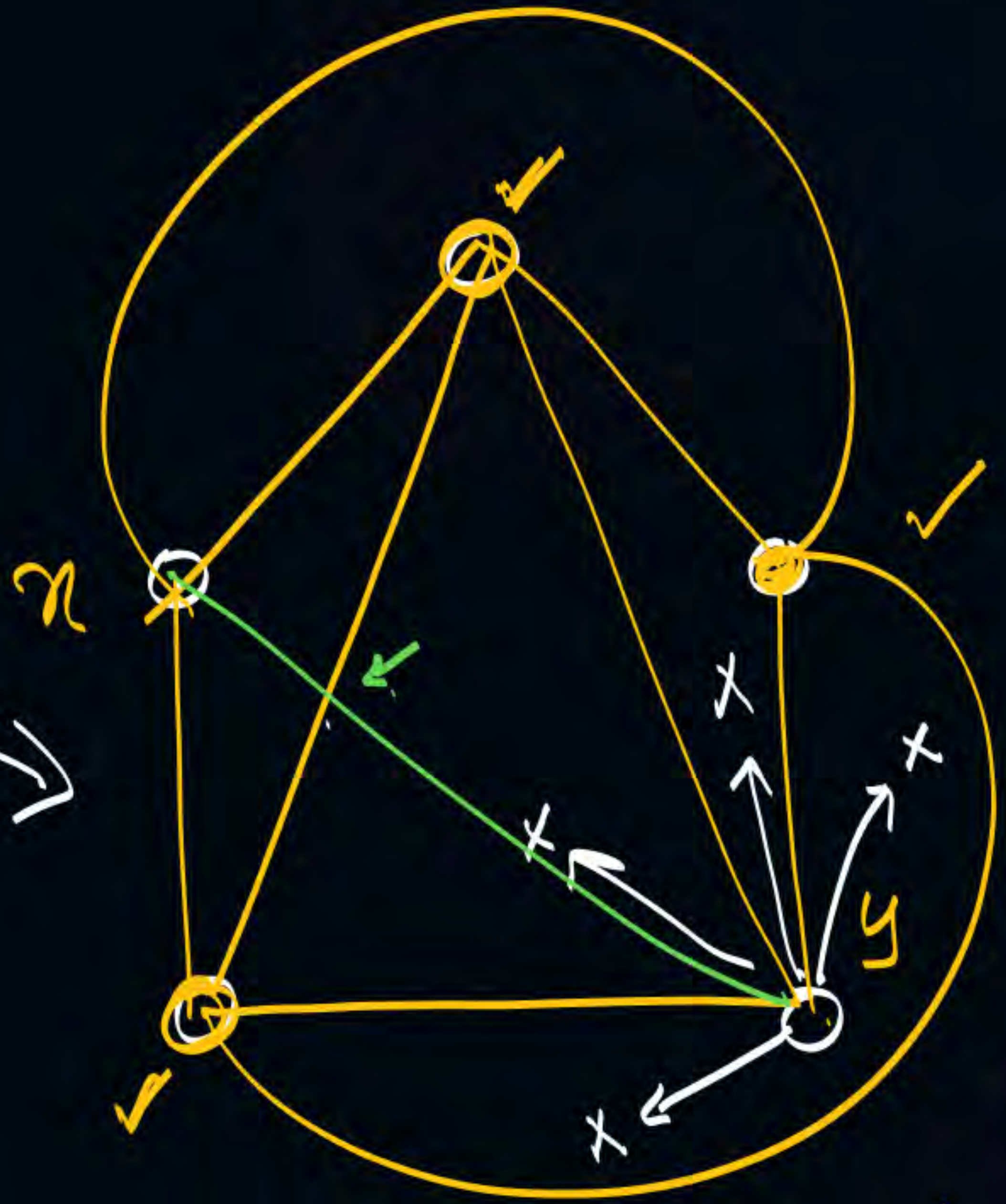
Planar Graph.



K_5

$\begin{cases} n = 5 \\ e = 10 \end{cases}$

nonplanar



non Planar

Planar

$\begin{cases} n = 5 \\ e = 9 \end{cases}$

$\begin{cases} K_5 \text{ nonplanar} \\ n = 5 \\ e = 10 \end{cases}$

$n=1$ • Pl ✓

planar:

$n=2$

$n=3$

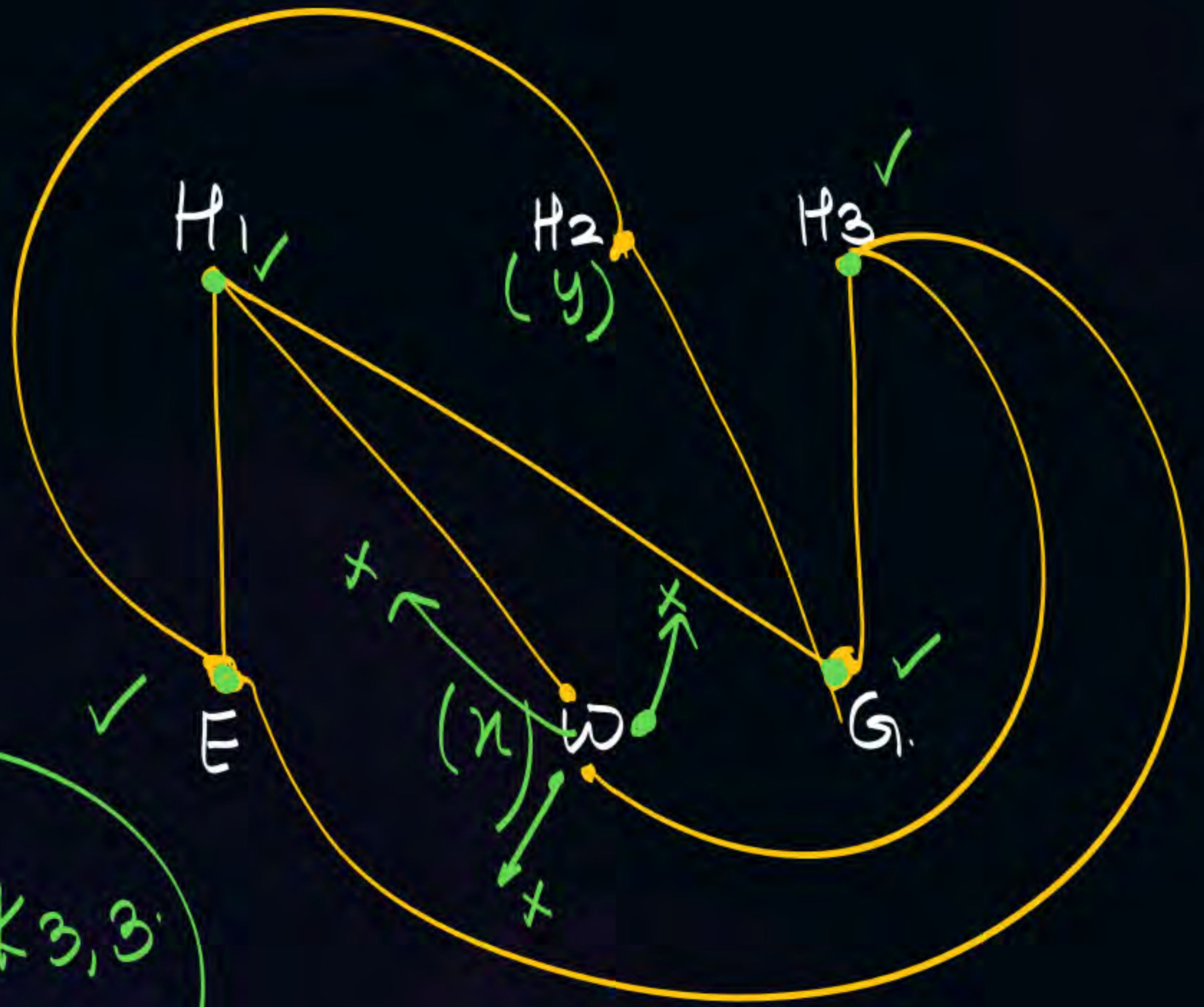
$n=4$

$n=5$ $e=0$ $e=1$ $e=2$ . . . $e=9$

Kurawtoski's

K_5 nonplanar

$n = 6$
$e = 8$
planar



$n = 6$
$e = 9$
$K_{3,3}$
non planar

$K_{3,3}$ $n = 6$ $e = 9$

E • • H1

W • • H2

G • • H3

* Both the graphs are Regular Graphs.

Removing single edge from both the graphs, will make them as planar Graph.

* Which nonplanar Graph is having min no. of vertices → K_5.

Which nonplanar Graph is having min no of edges → $K_{3,3}$

Nonplanar Graphs:

min.

K_5 $n = 5$ $e = 10$

min.

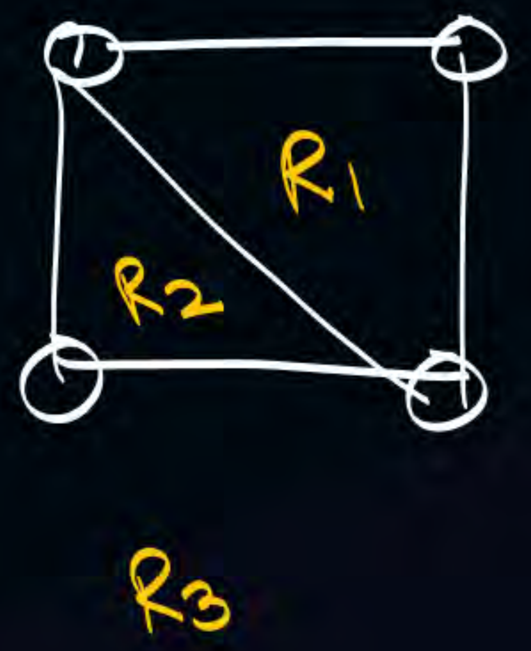
$K_{3,3}$ $n = 6$ $e = 9$

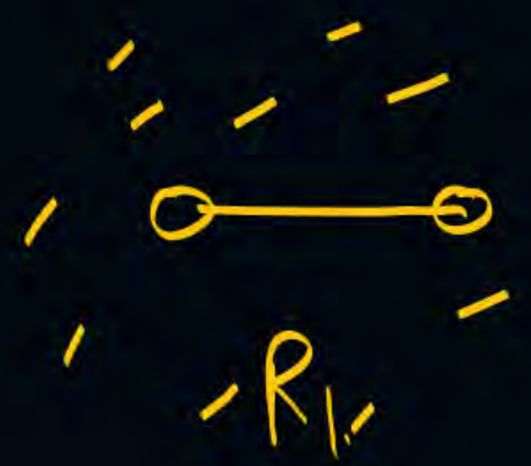
nonplanar

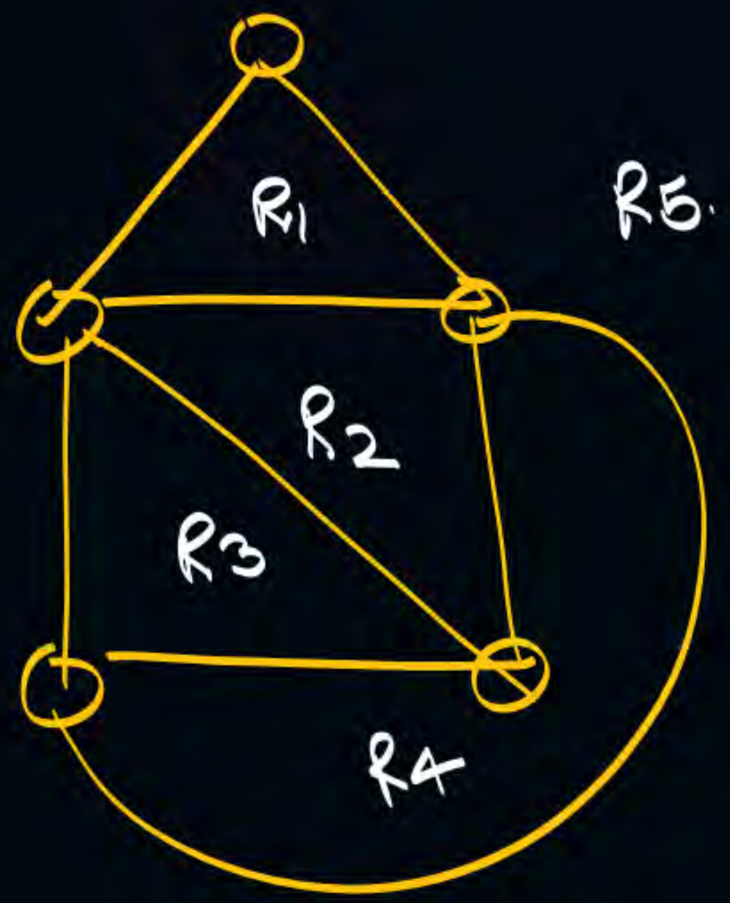
Embedding.

When we draw planar graph on a plane it creates Region / faces.

finite.	Infinite.
closed.	open.
bounded.	unbounded.


R1
R2
R3


R1


R1
R5
R2
R3
R4


R1
R2

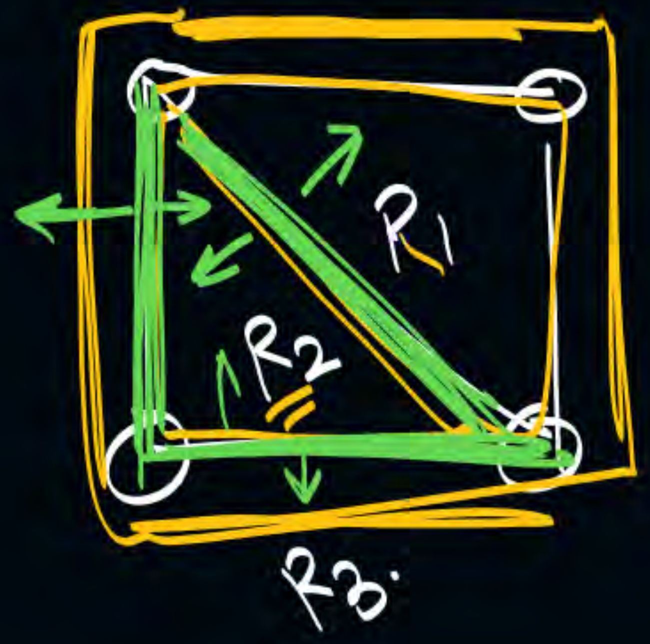
Degree of Regions: $d(R_i)$

$$\boxed{\sum d(R_i) = 2e}$$

no. of edges involved into Regions.

$deg(R_1) = 3$

$deg(R_2) = 3$



$deg(R_1) = 3$

$deg(R_2) = 3$

$deg(R_3) = 4$

$deg(R_1) + deg(R_2) + deg(R_3)$

$= 3 + 3 + 4$

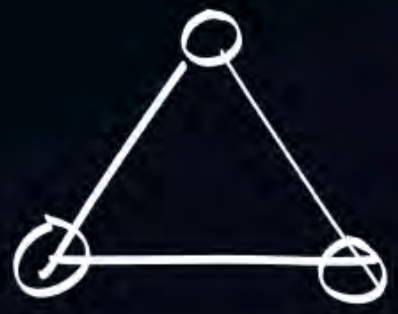
$= 10 = 2 \cdot 5$

no. of edges

Euler's formula (: PLANAR GRAPH)

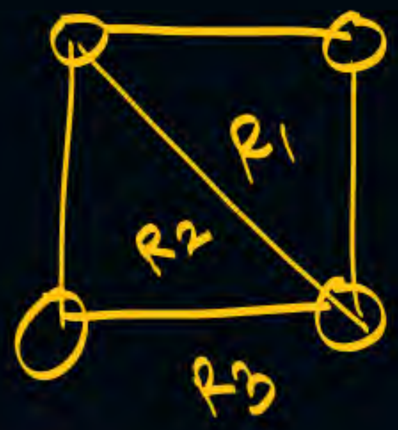
$$n - e + f = 2.$$

n = Total vertices.
e = Total edges.
f = Total Regions/ faces.



$n = 3$
$e = 3$

$$n - e + f = 2.$$
$$3 - 3 + f = 2$$
$$f = 2.$$



$n = 4$
$e = 5$

$$n - e + f = 2.$$
$$4 - 5 + f = 2.$$
$$f = 3$$

Planar Graph is having 10 vertices & 15 edges, what will be Total regions?

$n = 10$
$e = 15$

$n - e + f = 2.$

$10 - 15 + f = 2.$

$-5 + f = 2.$

$f = 7$

b) how many closed faces?

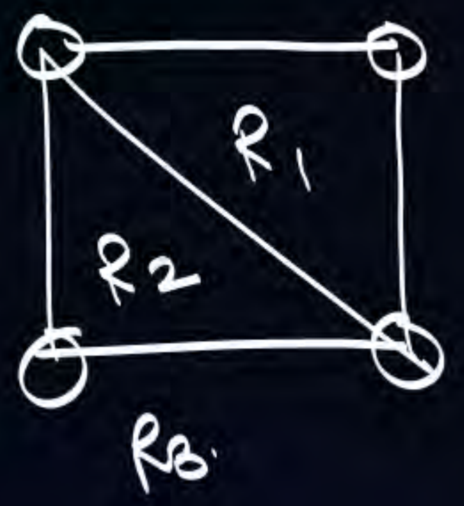
Closed = Total - Infinite.

$= 7 - 1 = \underline{\underline{6}}.$

$\sum d(R_i) = 2e$ $\quad\quad$ $n - e + f = 2.$

Planar Graph $(n \geq 3)$

Degree of each Region is made by atleast 3 edges.



$$\begin{cases} deg(R_1) = 3 \\ deg(R_2) = 3 \\ deg(R_3) = 4. \end{cases}$$

$$\begin{cases} deg(R_1) \geq 3 \\ deg(R_2) \geq 3 \\ deg(R_3) \geq 3. \end{cases}$$

$deg(R_1) + deg(R_2) + deg(R_3) \geq 3 + 3 + 3.$

$\sum d(R_i) \geq 3 \cdot 3$

$\sum d(R_i) \geq 3 \cdot f$ $\quad (f = 3)$

$$\boxed{2e \geq 3f}$$

$2e \geq 3f \qquad n-e+f=2.$

$f = 2+e-n.$

$2e \geq 3(2+e-n)$

$2e \geq 6+3e-3n.$

$3n-6 \geq 3e-2e.$

$3n-6 \geq e.$

Thm:

if G is planar then $e \leq 3n-6.$

but viceversa is not True.

if p then q

$p \rightarrow q$

meaning of viceversa

$q \rightarrow p$

$\Downarrow$

$p \rightarrow q$ is same as $\neg q \rightarrow \neg p$

if p.m exist then no. of vertices will be even

$p \rightarrow q$

$\Downarrow$

$\neg q \rightarrow \neg p$

p = p.m

q = no. of vertices even

if not (no. of vertices even) then not (p.m)

if no. of vertices are odd then p.m will not exist

Thm:

if $\underbrace{\text{G is planar}}_{p}$ Graph then $\underbrace{e \le 3n-6}_{q}$.

$p \to q$

$\Downarrow$

$\neg q \to \neg p$

if not ($e \le 3n-6$) then not (G is planar)

$\downarrow$

if $e > 3n-6$ then G is nonplanar Graph.

K5

$n = 5 \quad e = 10$

if $e > 3n-6$ then G is non planar.

$10 > 3(5)-6$

if. $10 > 9$. then G is non planar.

K3,3 $e = 9$ $n = 6$.

$$\boxed{\text{if } e > 3n-6 \text{ then } G \text{ is nonplanar.}}$$

$K_{3,3}$ $e = 9$ $n = 6$.

$9 > 3(6) - 6$. then G is nonplanar.

if $9 > 12$ then G is nonplanar.

F. $\longrightarrow$

Thm works
but example is wrong.

when Thm works:

$T \to T \equiv T$ ✓

$F \to T \equiv$ ✓

$F \to F \equiv T$ ✓

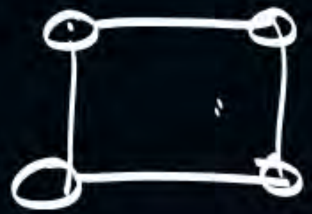
$F \longrightarrow$

when Thm does not works.

$T \to F \equiv$ thm does not work.

if p.m exist then no. of vertices will be even.

Case 1:



p.m → Even

T → T. ✓

F → True

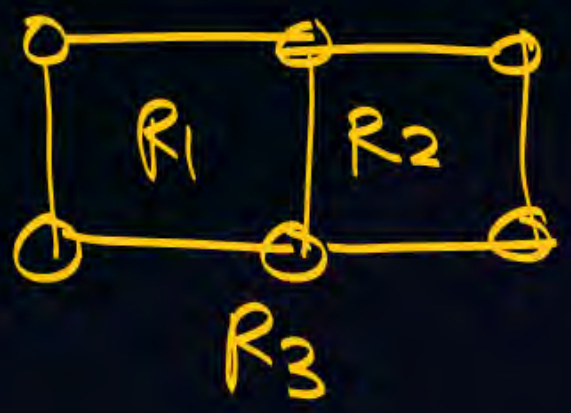
Case 2:

p.m → even.

F →

True

{ Planar Graph $(n \geq 4)$
{ Degree of each Region is made by atleast 4 edges.



$deg(R_1) \geq 4$.

$deg(R_2) \geq 4$.

$deg(R_3) \geq 4$.

$deg(R_1) + deg(R_2) + deg(R_3) \geq 4 + 4 + 4$.

$\sum d(R_i) \geq 4 \cdot 3$.

$\sum d(R_i) \geq 4 \cdot f$.

$2e \geq 4f$

$2e \geq 4f$ $n - e + f = 2$

$f = 2 + e - n$

$2e \geq 4(2 + e - n)$

$\div$ by 2

$e \geq 2(2 + e - n)$

$e \geq 4 + 2e - 2n$

$2n - 4 \geq e$

(atleast 4)

Thm: if G is planar then $e \leq 2n - 4$.

Thm: if $e > 2n - 4$ then G is nonplanar.

atleast 5

$2e \geq 5f$ $n - e + f = 2$.

$K_{3,3}$ $e = 9$ $n = 6$.

if $e > 2n - 4$ then G is nonplanar

$9 > 2(6) - 4$

$9 > 8$

True.

if G is planar then $e \le 3n-6$.

Thm:

$$\delta(G) \le \frac{2e}{n} \le \Delta(G) \le n-1.$$

Thm:

$\boxed{\text{if } G \text{ is planar then } \delta(G) \le 5}$

$$\delta(G) \le \frac{2e}{n}$$

$$\delta(G) \le \frac{2(3n-6)}{n}$$

$$\delta(G) \le \frac{6n-12}{n}$$

$$\delta(G) \le \frac{6n}{n} - \frac{12}{n}$$

$$\delta(G) \le 6 - \frac{12}{n}$$

$$\delta(G) \le 5$$

2 mins Summary

Topic One

Topic Two

Topic Three

Topic Four

Topic Five

Planarity

THANK - YOU